

▪ **Bus system:** Communication pathways that transport data between different components

In the context of IoT, von Neumann's architecture has played a pivotal role. IoT devices, at their heart, are miniaturised computers that interact with the physical world. They collect data through sensors, process this data, and then take actions or communicate findings. This sequence of operations—input (sensing), processing, and output (acting or communicating)—is a direct manifestation of von Neumann's principles.

However, this architecture does come with its set of limitations.

- **Bottleneck:** Often termed the 'von Neumann bottleneck', there's an inherent limitation in the speed at which data can be transferred between the CPU and memory. In real-time environments like IoT, this can cause latency issues.
- **Power consumption:** von Neumann architectures can be power-hungry, not ideal for IoT devices which often rely on battery power or energy harvesting.
- **Parallel processing:** The sequential nature of this architecture isn't always best suited for tasks that demand parallel processing, a feature that's becoming increasingly essential with the influx of large-scale data in IoT.

Despite these challenges, von Neumann's architecture has been the foundation of many breakthroughs in IoT. But as the scale and complexity of IoT systems grow, and the demand for real-time, efficient processing increases, there's a pressing need for architectures that can better handle these requirements. This sets the stage for the emergence of neuromorphic computing.

Transition to neuromorphic computing

In order to transcend the constraints of traditional computing systems, researchers have turned to an unlikely

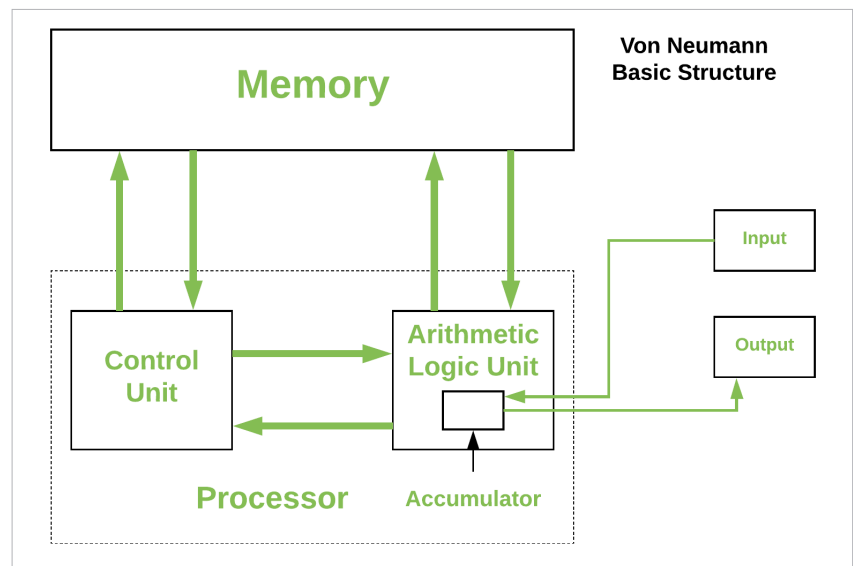


Figure 1: von Neumann architecture (Google Images)

yet profound source of inspiration: the human brain. The brain, with its billions of neurons and trillions of synaptic connections, operates in ways that no man-made machine can replicate — at least not yet. This realisation birthed the field of neuromorphic engineering, a domain that seeks to design algorithms, hardware, and systems inspired by the structure and functionality of biological neural systems. Figure 2 explores the architectural overview of neuromorphic computing.

Principles of neuromorphic computing

Parallelism over sequence: Unlike von Neumann architectures, which work predominantly in sequences, neuromorphic systems employ massively parallel operations, emulating the brain's vast network of simultaneously active neurons.

Event-driven processing: Neuromorphic systems often use event-driven models, such as spiking neural networks (SNNs). Instead of continuous computations, these models rely on discrete events or 'spikes', mirroring how neurons either fire or remain inactive.

Adaptability and learning: In traditional architectures, hardware and software are distinct entities. In contrast, neuromorphic systems blur these lines, allowing the hardware itself to adapt and learn, akin to the plasticity of biological synapses.

Memristors: The synaptic mimic

One of the groundbreaking components in neuromorphic engineering is the memristor. Memristors, whose resistance varies based on the history of currents passed through them, are seen as ideal components to represent the synaptic weights in artificial neural networks. Their capability to retain memory even when powered off mirrors the long-term potentiation and depression in biological synapses.

Why the shift matters for IoT

With billions of devices projected to be part of the IoT ecosystem, the demand is clear: low power consumption, real-time processing, and adaptive learning capabilities. Neuromorphic computing, with its energy efficiency and real-time adaptability, offers a promising avenue to address these needs. Devices can potentially learn and